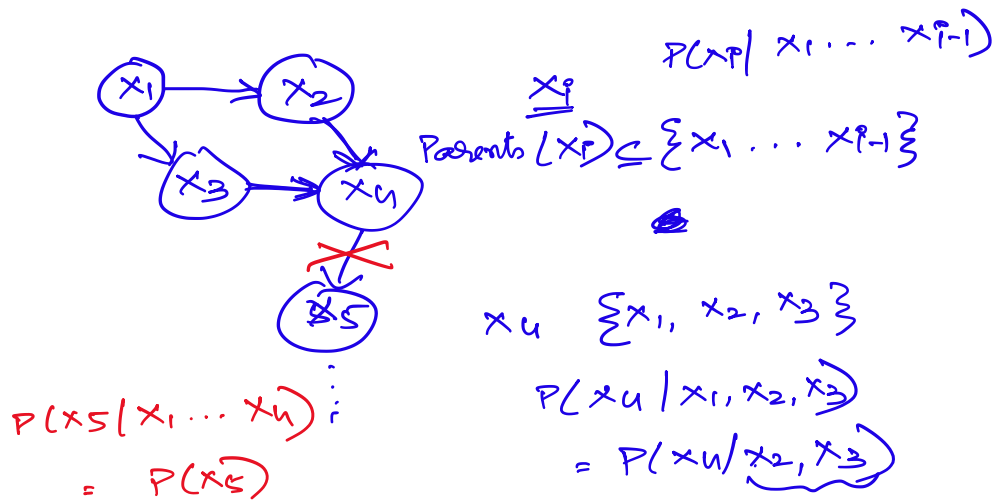


$$P(x_1 \dots x_N) = P(x_1) P(x_2 | x_1) P(x_3 | x_1, x_2) \dots P(x_N | x_1 \dots x_{N-1})$$

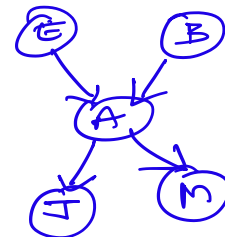
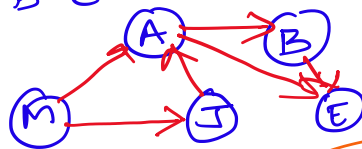
$$P(x_i | x_1 \dots x_{i-1}) = P(x_i | \text{Parents}(x_i))$$



$x_1 \dots x_N$

$$P(x_i | x_1 \dots x_{i-1}) = P(x_i | \text{Parents}(x_i))$$

M I A B E



$I \quad P(I|M) \neq P(I) ? \quad \frac{P(A, I, M, E, B) \cdot P(M) P(B|A)}{P(A|I, M) P(B|A) P(E|A, B)}$

$A \quad \left[\begin{aligned} P(A|I, M) &= P(A) ? \\ &= P(A|I) ? \\ &= P(A|M) ? \\ &= P(A|I, M) \end{aligned} \right]$

$\text{Parents}(A) = \{ \}$
 $= \{ M \}$
 $= \{ I \}$
 $= \{ M, I \}$

$B \quad \left[\begin{aligned} P(B|A, I, M) &= P(B) \times \\ &\frac{P(B|A)}{P(B|A, I)} \end{aligned} \right] \checkmark$

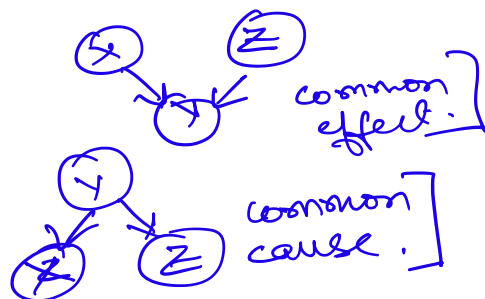
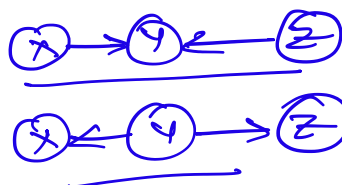
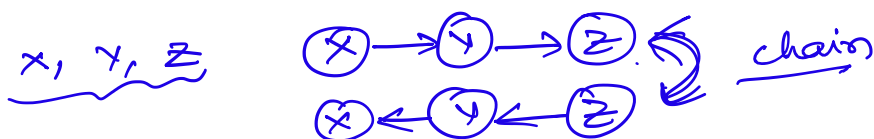
$P(B, E, A, I, M) \cdot P(B) P(E)$
 $\frac{P(B|A, E)}{P(I|A) P(M|A)}$

$$\begin{array}{l} P(B|A, M) \\ P(B|S, M) \\ \hline P(B|A, S, M) \end{array} \Bigg| X$$

$$E \quad P(E|B, A, S, M) = \frac{P(E|A)}{P(E|A, B)}$$

$$X \perp Y$$

$$P(X|Y) = P(X) \equiv P(Y|X) = P(Y) \equiv P(X, Y) = P(X) P(Y)$$



causal chain

$$\boxed{P(X, Y, Z) = P(X) P(Y|X) P(Z|Y)}$$

$$X \perp Z$$

$$P(Z|X) = P(Z)$$

$$\begin{aligned} P(Z|X) &= \frac{P(X, Z)}{P(X)} \\ &= \sum_Y P(X, Y, Z) \end{aligned}$$

$$P(Z=z|X=x) = \frac{\sum_Y P(X=x, Y, Z=z)}{P(X=x)}$$

$$= \frac{P(X=x, Y=y, Z=z) + P(X=x, Y=\neg y, Z=\neg z)}{P(X=x)}$$

$$= \frac{0.8}{0.8} \cdot \frac{1}{1} \cdot \frac{1}{1} + \dots$$

$$\begin{aligned} P(X=x) &= 0.8 \\ P(X=\neg x) &= 0.2 \end{aligned}$$

$$P(Y|X)$$

X	P(Y=y X)	P(Y=\neg y X)
1	1	0
0	0	1

$$P(Z|Y)$$

Y	P(Z=z Y)	P(Z=\neg z Y)
1	1	0
0	0	1

$$\begin{aligned}
 &= P(X=x) P(Y=y|X=x) P(Z=z|Y=y) \\
 &= \frac{0.8}{0.8} = 1
 \end{aligned}$$

$$\begin{aligned}
 P(Z) &= \sum_{x,y} P(X=x, Y=y, Z=z) \\
 &= \sum_{x,y} P(X=x) P(Y=y|X=x) P(Z=z|Y=y)
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{P(X=x) P(Y=y|X=x) P(Z=z|Y=y) +}{P(X=x) P(Y=y|X=x) P(Z=z|Y=y) +} \\
 &\quad P(X=\neg x) P(Y=y|X=\neg x) P(Z=z|Y=y) + \\
 &\quad P(X=\neg x) P(Y=\neg y|X=\neg x) P(Z=z|Y=\neg y) \\
 &= 0.8
 \end{aligned}$$

$$\underline{P(X, Z) = P(X) P(Z)}$$

$$\begin{aligned}
 P(Z|X=x, Y=y) &= P(Z|Y=y) \checkmark \\
 Z \perp X | Y
 \end{aligned}$$

Bayes Network

$x_1, \dots, x_N \quad x_i$

Parents (x_i) $\subseteq \{x_1, \dots, x_{i-1}\}$

$$P(x_i | x_1, \dots, x_{i-1}) = P(x_i | \text{Parents}(x_i))$$

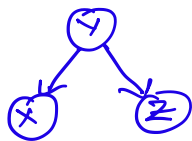


$$P(x, y, z) = P(x) P(y|x) P(z|y)$$

$$x \perp z | y = z \perp x | y$$

$$P(z|y, x) = P(z|y)$$

$$\begin{aligned}
 P(z|y, x) &= \frac{P(x, y, z)}{P(x, y)} = \frac{P(x) P(y|x) P(z|y)}{\sum_z P(x, y, z)} \\
 &= \frac{P(x) P(y|x) P(z|y)}{P(x, y, z) + P(x, y, \neg z)} \\
 &= \frac{P(x) P(y|x) P(z|y)}{P(x) P(y|x) P(z|y) + P(x) P(y|x) P(\neg z|y)} \\
 &= \frac{P(x) P(y|x) P(z|y)}{P(x) P(y|x) [P(z|y) + P(\neg z|y)]} \\
 &= P(z|y)
 \end{aligned}$$



$$P(x, y, z) = P(y) P(x|y) P(z|y)$$

$$P(y) = 0.8 \quad P(\neg y) = 0.2$$

y	x	P(x y)	y	z	P(z y)
1	1	1	1	1	1
0	0	1	0	0	1

$x \perp z$? $\times$

$$P(x, z) = P(x) P(z) \equiv P(x|z) = P(x) \equiv P(z|x) = P(z)$$

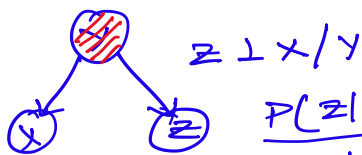
$$P(x, z) = \sum_y P(x, y, z)$$

$$\begin{aligned}
 &= P(x, y, z) + P(x, \neg y, z) \\
 &= P(y) P(x|y) P(z|y) + P(\neg y) P(x|\neg y) P(z|\neg y) \\
 &= \underline{0.8}
 \end{aligned}$$

$$P(x) = \sum_{y, z} P(x, y, z)$$

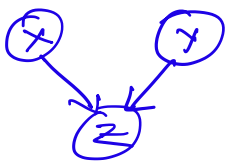
$$\begin{aligned}
 &= P(x, y, z) + P(x, y, \neg z) + P(x, \neg y, z) + P(x, \neg y, \neg z) \\
 &= P(y) P(x|y) [P(z|y) + P(\neg z|y)] + \\
 &\quad P(\neg y) P(x|\neg y) [P(z|\neg y) + P(\neg z|\neg y)] \\
 &= 0.8 \cdot 1 + 0 = 0.8
 \end{aligned}$$

$$P(z) = 0.8$$



$$\begin{aligned}
 &Z \perp X/Y \\
 &\frac{P(Z|Y, X)}{P(X|Y, Z)} = \frac{P(Z|Y)}{P(X|Y)} \\
 &P(X, Z|Y) = P(X|Y) P(Z|Y)
 \end{aligned}$$

$$\begin{aligned}
 P(Z|Y, X) &= \frac{P(X, Y, Z)}{P(X, Y)} \\
 &= \frac{P(Y) P(X|Y) P(Z|Y)}{P(Y) P(X|Y) P(Z|Y) + P(\neg Y) P(X|\neg Y) P(Z|\neg Y)} \\
 &= \frac{P(Y) P(X|Y) P(Z|Y)}{P(Y) P(X|Y) [P(Z|Y) + P(\neg Z|Y)]} \\
 &= P(Z|Y)
 \end{aligned}$$



$$P(x, y, z) = P(x) P(y) P(z|x, y)$$

$$X \perp Y$$

$$P(x, y) = P(x) P(y) ?$$

$$\hookrightarrow \sum_z P(x, y, z)$$

$$\begin{aligned}
 &= P(x, y, z) + P(x, y, \neg z) \\
 &= P(x) P(y) [P(z|x, y) + P(\neg z|x, y)] \\
 &= P(x) P(y)
 \end{aligned}$$

$$x \perp y | z?$$

$$P(x|z, y) = P(x|z)?$$

$$\sum_{e, a} P(B, e, a, j, m)$$

$$\sum_{e, a} P(B) P(e) P(a|B, e) P(j|a) P(m|a)$$

$$\begin{aligned}
 &P(B) P(e) P(a|B, e) P(j|a) P(m|a) + \\
 &P(B) P(e) P(a|B, e) P(j|\neg a) P(m|\neg a) + \\
 &P(B) P(\neg e) P(a|B, \neg e) P(j|a) P(m|a) + \\
 &P(B) P(\neg e) P(a|B, \neg e) P(j|\neg a) P(m|\neg a)
 \end{aligned}$$

